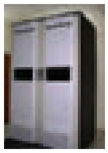
Your very own Sony supercomputer
Scientists in the US have built a supercomputer using nothing more than the computing power of PS3 consoles! Check out **http://www.ps3cluster.umassd.edu**



Home grown muscle
India's first supercomputer was the PARAM 8000. PARAM stands for Parallel Machine. The computer was developed by the government run C-DAC in 1991

Super-jargon busting

"FLOPS" or Floating point Operations Per Second (fractional values in short) is the standard measure that's used to gauge the performance of supercomputers.Commonly used with SI prefixes such as tera- (10^{12}) or peta- (10^{15}),teraflops and petaflops can be used in shorthand as TFLOPS and PFLOPS respectively. "Petascale" supercomputers can process one quadrillion (10^{15} or 1000 trillion) FLOPS. Exascale is computing performance in the exaflops range.An exaflop is one quintillion (10^{18}) FLOPS, or one million teraflops!

Be-aware the "Beowulf"

Open Source Cluster Application Resources (OSCAR) is a Linux-based software installation for high-performance cluster computing. OSCAR allows users to install a Beowulf type high performance computing cluster. Beowulf Clusters are scalable performance clusters based on commodity hardware, on a private system network, with an open source software (Linux) infrastructure. The OSCAR project aims to support as many different Linux distributions as possible. Supported Linux distributions as mentioned on the OSCAR website are Debian, Fedora, OpenSUSE, Red Hat Enterprise Linux and Ubuntu (http://bit.ly/p4vur).

Example

For instance, if you were to create a cluster at home using one dual core AMD system running Windows 7 and one single core Intel system running Ubuntu (yes it is possible since clusters are platform independant), theoretically, the total resource pool available to the computer will be 3 cores. But as explained, bottlenecks are caused due to varying factors such as limited bandwidth over the LAN and varying hardware (slower RAM, processors on other nodes etc.).

cross cable for two PCs that are connected directly, or regular patch cables to hook them all up to a switch. The Primary acts as a DHCP server, so you're going to have to make sure that you don't mess with any other DHCP servers running on the network (like at work, or in a college). For those who don't know, a Dynamic Host Configuration Protocol server is one that gives PCs on the network the IP configuration information such as their own IP, the subnet mask and gateway information.

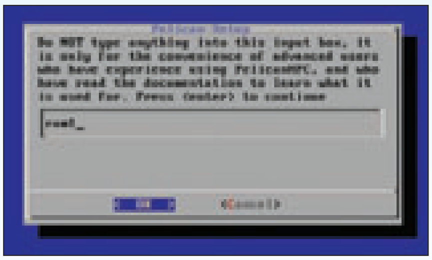
You could either copy the ISO file off the Mindware DVD on your local machine and boot it up using virtualbox or burn the image onto a CD (like we did). The advantage is that booting the live image will offer better performance than a virtual machine.

Select "Live" to boot the image into RAM

Let's begin

When the PC boots up, press [`Enter`], or whatever you're asked to, in order to boot from the CD. Select "Live" when PelicanHPC asks you for a boot method.

You will eventually be greeted by a screen that gives you the option of booting from your permanent storage. If you wish to do so, you'll have to assign drives accordingly (sda1, sda3 etc.). For our article, we're sticking to "ram1", so choose that. On the next screen, you will probably want to move the cursor over to "Yes" and press enter unless you are running a customised version of PelicanHPC. Customised versions are relatively easy to create, and you should read the PDF to learn how you can bundle all the packages you find necessary



Boot into ram1 to prevent any damage to the OS on the primary PC

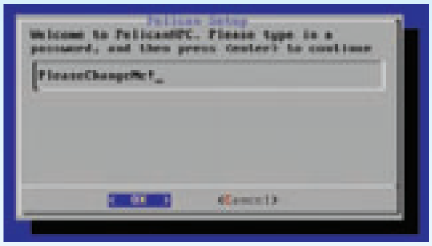
into an image so that all packages are consistent across your cluster.

Ok, almost done! Now change the default password and enter your own password. Hit [`Enter`].

Now you are in the PelicanHPC terminal. Enter your username as "user" (no quotes) and then type the password you entered. You are now logged in. You can now choose to enter the graphical user

REMEMBER!

The administration of your cluster will require you to keep 2 commands in mind. They are "pelican_setup" for configuring the frontend as a server (so that all other compute nodes can LAN boot off the frontend) and "pelican_restart_hpc" which is used to add/remove nodes from your cluster after initial setup. These commands need to be run in terminal.



Change the password

interface by typing "startx" or continue to use the console. The console is the default mode in PelicanHPC to avoid problems with unusual graphics hardware.

If you boot into the Xfce graphical interface, you will have to start terminal and then proceed.

Now, time to setup your cluster. In terminal type "pelican_setup".

When you arrive at the screen where the instructions state "Time to bring the compute nodes into the cluster", hit [`Enter`] on "Yes" and boot up the compute nodes making sure that the computers are connected via ethernet cable(s) and that their first boot device is set to Network.

Tip: For those of you having trouble netbooting your old PC's, check out http://www.rom-o-matic.net for help on how to get your netboot up and running.

The compute nodes will netboot and stop at prompt.

Now, keep hitting "No" till the complete count of the compute nodes is available on the screen. Once all compute nodes are connected and accounted for, hit "Yes". Your compute cluster is now successfully up and running!

Tip: When you are done with the compute nodes, hit `Alt+Ctrl+Del` to restart your PC.